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Research article

Sighting Common Snapping Turtle Chelydra serpentine, Linnaeus 1758, at Silver Spring, Maryland, USA and its account with other aspects

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ABSTRACT

Present communication deals with the sighting of Chelydra serpentina, the Common Snapping Turtle at Silver Spring, Maryland (USA) and reported here for the record. Its synonymy, diagnostic features, distribution, food & feeding, breeding, nesting migration, behaviour, conservation status, threats and control measures have been provided. Variations in length of carapace and plastron, body weight and maturity-size as per various records have also been given.

Keywords

Sighting Snapping Turtle at Silver Spring, Maryland.

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INTRODUCTION

Chelydra serpentina, the Common Snapping turtle is widely distributed in United States and Canada and good work on its various aspects has been done by various workers (White & Murphy, 1973; Obbard & Brooks, 1980; Galbraith et al., 1989; Haxton, 2000; Wilson & Burnie, 2001; Ewert et al., 2005; Medina, 2006; Rohdin et al., 2010; Tekiela, 2014; Kiley, 2018 and others) but no sighting / occurrence record from Silver Spring (Maryland), in particular, is available and hence provided here for general information.

STUDY SITE: SILVER SPRING

Physiography: Silver Spring is located at Lat. 39o0'8.79" N and Long. 77o1'14.85"W and at elevation of 106 m above sea level in south-eastern Montgomery County, Maryland, United States, near Washington, D.C. It covers a total area of 20.5 km2 / 7.92 miles2 which is all land with green pastures, plants, ponds and creeks (Rock Creek, Sligo Creek, Long Branch, and Northwest Branch). Its name is derived from a local spring, the bottom of which sparkled with flakes of mica.

Climate: The summers warm and humid, winters very cold and windy and partly cloudy year round. Over the course of the year, average temperature typically varies from 25° F / -3.8° C (in January) to 88° F /31° C (in July) and is rarely below 15° F / -9.4o C or above 95° F / 35o C. It gets 44 in / 1118 mm of rain, on average and 14 in / 256 mm of snow per year.

Flora (Maryland): Trees: Liquidambar styraciflua, the Sweet Gum (Altingiaceae); Quercus alba, the White Oak; Q. falcata, the Southern Red Oak; Q. montana, the Chestnut Oak; Q. rubra, the Northern Red Oak; Q. velutina, the Black Oak; Fagus grandifolia, the American Beech; (Fagaceae); Carya

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Figure 1 *Chelydra serpentine*, the Common Snapping Turtle (Courtesy: Dr. Syed Rafat Husain)

glabra, the the Pignut Hickory (Juglandaceae); Sassafras albidum, the Silky Sassafras (Lauraceae); Liriodendron tulipifera, the Tulip Tree (Magnoliaceae); Nyssa sylvatica, the Black Gum (Nyssaceae); Fraxinus americana, the White Ash (Oleaceae); Pinus strobus, the Eastern White Pine; P. taeda, the Loblolly Pine; P. virginiana, The Virginia Pine; Tasuga canadensis, the Eastern Hemlock (Pinaceae); Platanus occidentalis, the American Sycamore (Platanaceae); Acer rubrum, the Red Maple, A. saccharinum, the Sillver Maple; A. saccharum, the Sugar Maple (Sapindaceae).

Bushes & Herbs: Campsis radicans, the Trumpet Vine (Bignoniaceae); Eutrochium dubium, the Coastal Plain Joe Pye weed and E. purpureum, the Sweet Joe Pye Weed (Asteraceae); Asclepias syriaca, the Butterfly Flower (Apocynaceae); Polygonatum biflorum, the Solomon's Seal (Asparagaceae); Solidago missouriensis, the Missouri Golden-rod (Asteraceae); Dianthus spp. (Caryophtllaceae); Kalmia latifolia, the Calico-bush or Spoon-wood (Ericaceae); Baptisia australis, the Blue False Indigo (Fabaceae); Monarda didyma, the Crimson Bee-balm (Lamiaceae); Actaea racemosa, the Black Cohosh and Anemone canadensis, the Canada Anemone (Ranunculaceae); Parthenocissus quinquefolia, the Virginia Creeper (Vitaceae).

Chelydra serpentina (Linnaeus, 1758)

Synonymy:

Tesudo serpentina Linnaeus, 1758. Syst. Nat., 10 Edn.:199 (type-locality: in Calidis regionibus); Shaw & Nodder, 1813. The Naturalist's Miscellany, 24: pl. 1062.

Testudo serpentaria Wiedemann, 1802. Archiv fur Zoologie und Zootomie, 2 (2): 191 (nomen novum).

Chelydra serpentina, Schweigger, 1812. Konigsberg. Arch. Naturwiss. Mathem., 1: 271-368, 406-458; Gunther, 1885. Reptilia and Batrachia. Biologia Centrali-Americana: 10; Stebbins, 1985. A Field Guide to Western Reptiles and Amphibians (2nd edn.): 97; Conant & Collins, 1991. A Field Guide to Western Reptiles and Amphibians of Eastern/Central North America (3rd ed.): 41; van Dijk et al., 2014. Conservation Biology of Freshwater Turtles and Tortoises. Chelonian Research Monographs, No. 5: 340.

Chelydra lacertina, Schweigger, 1812. Konisnerg. Arch. Naturwiss. Mathem., 1: 293 (senior homonym).

Chelonura serpentina, Say, 1824. J. Acad. Nat. Sci. Philadelphia, 4 (2): 206.

Emysaura serpentina, Dumeril & Bibron, 1835. Erpetologie Generale ou Histoir Naturelle Complete des Reptiles, 2: 350.
Chelydra emarginata Agassiz, 1857. Contributions to the Natural History of the United States: 417.

Macroclemys lacertina, Rhoads, 1895. Proc. Acad. Nat. Sci. Philadelphia, 1895: 381; Largelamb, 1907. National Geographic, 18 (6): 413-419.

Devisia mythodes Ogilby, 1905. Proc. Roy. Soc. Queensland, 19 (1): 11.

Chelydra serpentina serpentina, Stejneger, 1914. Copeia, 1 (6): 3-4.

Chelydra laticarinata Hay, 1916. Annual Report of the Florida State Geological Survey, 8: 39-76. (nomen suppressum), ICZN 1989).

Chelydra sclupta Hay, 1916. Annual Report of the Florida State Geological Survey, 8: 39-76. (nomen suppressum, ICZN 1989).

Chelydra serpentina osceola, Stejneger, 1918. Proc. Biol. Sci. Washington, 31: 89 (nomen conservandum, ICZN 1989).

Common Names: Common Snapping Turtle, North American Snapping Turtle and Snapping Turtle.

Classification: Class Reptilia, order Testudines Batsch, 1788, suborder Cryptodira Cope1868, family Chelydridae Gray, 1831, genus Chelydra Schweigger, 1812.

Etymology: Chelydra derived from Greek word 'kheludros' meaning an amphibious serpent, a kind of tortoise and serpentina from Latin word 'serpens' meaning snake, referring to its long tail, bite and often aggressive nature (reptile-database).

Sighting: Recently sighted at U. S. Food and Drug Administration White Oak Campus (Coordinates 39o02'07" N, 76o58'59" W), 10903 New Hampshire Ave, Silver Spring, Montgomery County, Maryland 20993 (USA).

Diagnostic Features:

Morphology: Rugged muscular built-body; head large with knobs of skin around mouth, jaws powerful and sharp-hooked; hard beak with rough cutting edge; neck long and thick; carapace large, ridged, with three knobby keels and posterior margin strongly serrated; last four marginals on each side of midline strongly pointed; juveniles with three prominent serrated keels/ridges on their carapace, tubercles on tail develop with age; plastron hinge-less, much short, reduced to a cross-sturt; plastral bridges connected to carapace by a ligament (not fused as in other turtles); tail long with lateral spines; feet webbed with long sharp claws.

Colouration: Carapace dark green, brown or sometimes black; lower shell light yellow or whitish. Carapace, plastron and skin of hatchlings black, white spots on marginals and on margin of plastron (Virginia herpetological society).

Carapace Length: Estimated 45-60 cm (present); female 24.9 cm (Galbraith et al., 1989); males larger than females (Brooks et al., 1991); av. 20.3-36 cm, max. 46.5 and 49.4 cm (Conant & Collins, 1991; Virginia herpetological society); average 28.5 cm (Iverson et al., 1997); female 26.4-280 cm, male at 27.6 cm (Haxton, 2000); in adulthood nearly 50 cm, commonly 25-47 cm (Wilson & Burnie, 2001); male 30-48 cm, female 25-38 cm (Tekiela, 2014); up to 50 cm (nature.ca); av. 20-30 cm (portal.ct.gov).

Plastron Length: 14.5 cm (White & Murphy, 1973); average 22.5 cm (Iverson et al., 1997); 32.4 cm (Virginia herpetological society).

Weight: In excess of 10 kg (Brooks et al., 1991); about 6 kg (Iverson et al., 1997); up to 16 kg / 35 lb (Steyermark et al., 2008); some individuals can weigh 23 kg or more (Tekiela, 2014); 15 kg (nature.ca); 4.5-16 kg /10-35 lb (portal.ct.gov); 9-18 kg / 20-40 lb (tortoisetown.com); 4.5-16 kg, 23.13, heaviest 34 kg, in captivity up to 39 kg in captivity (Virginia herpetological society; Wikipedia).

Sexual Dimorphism: Adult male larger than female including head; plastron of some males slightly concave; pre-cloacal distance more in male than in female (Virginia herpetological society). White & Murphy (1973) also found sexual difference in pre-cloacal tail length having values significantly more in males than in females.

Altitudinal Range: Up to 2,000 m (IUCN Red List).

Distribution: Eastern United States to the Rocky Mountains, from southern Canada to the Gulf of Mexico and into Central America.

Maryland: Silver Spring (Food and Drug Administration, White Oak Campus) (present sighting record).

Elsewhere: Canada (west to Manitoba, New Brunswick, Nova Scotia, Ontario, S Quebec and Saskatchewan, SE Alberta), United States (Alabama, Arkansas, Colorado, Connecticut, Delaware, Florida, Georgia, Illinois, Indiana, Kansas, Kentucky, Louisiana, Iowa, Maine, Massachusetts, Michigan, Minnesota, Mississippi, Missouri, Montana, Nebraska, New Hampshire, New Jersey, New Mexico, New York, North Carolina, North Dakota, Ohio, Oklahoma, Pennsylvania, Rhode Island, South Carolina, South Dakota, Tennessee, Texas, Vermont, Virginia, West Virginia, Wisconsin and Wyoming).

Introduced: China, Chiba Prefecture, Italy, Japan (Honshu, main island), Rome, Taiwan, United States (Arizona, California, Nevada, Oregon and Washington).

Habitat: Highly aquatic in Inland Waters (ponds, pools, lakes, wetlands, swamps, marshes, bogs, creeks, streams, rivers etc.) and coastal areas with aquatic vegetation and cover in the form of stumps, overhanging ledges, musk-rat lodges / burrows and sandy substrates.

Food and Feeding: Basically carnivorous, feeding on arthropods, snails, fishes, frogs & toads, lizards, birds, small mammals (musk-rats etc.), carrions (decaying or dead animals, like dead fish and drowned mammals) but also take aquatic vegetable matter.

Nesting Migration: Obbard & Brooks (1980) studied the movements of female using radio-telemetry and mark-recapture techniques during annual migration to nesting sites and observed that most females move downstream but only a few towards upstream, some travelling more than 0.5 km over land from one place to another in search of suitable site with a maximum 16 km distance.

Breeding: Oviparous. Mating takes places in water. Female matures sexually at around 12 years, lays up to white 13-48 eggs (Ash, 1951); up to 100 eggs (Steyermark et al., 2008) / 25-80 eggs (Wikipedia) / 20-40 eggs (portal.ct.gov); 7-55 eggs (Virginia herpetological society) in flask-shaped cavity dug by female in mud on land once in a year during mid May -mid June and after covering the eggs with mud or sand returns to water. Hatching takes place in 60-130 days in spring season (September or early October) and new born fend themselves. The females mature at the carapace length of 26.4-280 cm and the male at 27.6 cm and over carapace length during nesting period, May-June, peak in mid June (Haxton, 2000). Galbraith et al. (1989) found the females laying eggs at 24.9 cm carapace length, corresponding to an age of 18 years. However, Haxton (2000) didn't find mature female at lesser than 26 cm carapace length and estimated 13 year of age and found youngest mature male with carapace length 27.6 cm and estimated 14-year age. White & Murphy (1973) found both sexes becoming mature at about 14.5 cm plastron length. Ewert et al. (2005) studied the sex ratios and observed that they are female-biased at cool temperatures, male-biased at moderate temperatures and only females are produced at warm temperatures. They also found that at high latitudes females nest in exposed warm locations and at low latitudes in shaded places which prevent from overheating. Such nest site selection renders high latitude nests warmer than low latitude nests, even if only intermittently.

Life-span: 30-40 years (nature.ca); 80-100 years (tortoisetown.com); up to 40 years or more (portal.ct.gov).

Behaviour / Nature: Being generally nocturnal and more active at night, foraging and pursuing their food. Quite docile and avoid confrontation but can deliver a powerful bite in defence when in danger on land (Kiley, 2018) but in water slip quietly away from any disturbance. They can tolerate brackish water well.

Bite and Treatment: Normally they don't bite unless scared or injured by some way. However, if bitten, the wound should have cleaned and get treated by antibiotics as they carry salmonella.

Hibernation: Hibernate during winter by getting bury into underwater mud. During this period, they do not breathe in normal way but getting oxygen through membranes of mouth and throat.

Conservation Status: IUCN Red List- Least Concern. In Canada listed as 'Special Concern' in the Species at Risk Act in 2011. Was declared the state reptile of New York by New York Legislature in 2006 (Medina, 2006).

CITES: Appendix III (USA).

Threats: Road-kill and other mortality; Raccoon and Bald eagle predation, pet trade; habitat loss and degradation, water pollution (by way of pesticides, industrial chemicals and heavy metals), consumption as food; predation on juveniles by large wading birds, large fishes and some snakes; eggs in nests eaten by Raccoons, skunks, foxes, crows and snakes; leech infestation.

Control Measures: (i) Avoiding consumption for food, also in view of presence of toxic chemicals in its flesh and fat. (ii) Placing caution signs on road sides where they cross for breeding. (iii) Making speed-breakers on roads of their crossing. (iv) Controlled harvesting for proper population management.

Remarks: Presently, no subspecies of the common snapping turtle are recognized (Rhodin et al., 2010). The former Florida subspecies osceola is currently considered a synonym of serpentina, while the other former subspecies Chelydra rossignonii (Bocourt, 1868), the Central American Snapping Turtle (from Belize, Guatemala and Mexico) and Chelydra acutirostris Peters, 1862 , the South American Snapping Turtle (from Colombia, Costa Rica, Ecuador, Honduras, Nicaragua and Panama) are both recognized as full species (Rhodin et al., 2010; van Dijk et al., 2014; reptile-database.org).

Feuer (1971) reported intergradations between Chelydra serpentina serpentina and Chelydra serpentina osceola, indicating incomplete separation and therefore subspecies status. This is confirmed by Shaffer et al. (in Steyermark et al., 2008) who reject osceola as a distinguishable subspecies.

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